PERFORMANCE CHARECTERISTICS OF GEAR OIL PUMP

AIM:-

To study the performance of a gear oil pump and to plot the following graphs.

1. Head v/s Discharge
2. Input Power v/s Discharge
3. Output Power v/s Discharge
4. Efficiency v/s Discharge

APPARATUS USED

1. Gear oil pump with suction and delivery pipes and necessary valves.
2. A measuring tank fitted with a piezometer tube and drain valve.
3. An energy meter fitted in the apparatus.
4. Stop watch.

PRINCIPLE

Discharge $Q = \frac{AR}{t}\frac{m^3}{s}$

Where

A = area of collecting tank in m^2 (0.25×0.25 m^2)

R = rise of Oil level in the collecting in m

t = time in seconds for raise of oil level 'R'

Efficiency of the pump, η $= \frac{Output}{Input} \times 100$ %

Out put power, OP $= \frac{\rho_o g Q H}{1000}$ kW

Where

ρ_o = density of oil in kg/m^3 (800 kg/m^3)

g = acceleration due to gravity in m/sec^2

Q = discharge of oil in m^3/sec

H = total head in m of oil

Total head, $H = H_d + H_s + l_d$

H_d= Delivery Pressure head in meters of oil

H_s = Suction Pressure head in meters of oil

l_d=Level difference between delivery and suction pressure headsin meters of oil

$$\textbf{Input power, } IP = \frac{n3600\eta_m}{TE_m} \text{ kW}$$

Where

n	= Number of revolution of energy meter.
T	= Time for 'n' revolution of energy meter in sec.
E_m	= Energy meter constant in rev/kWh
η_m	= Efficiency of the motor = 80%

PRECATIONS

1. Never run the pump without oil in it as this would cause damage to stuffing box, bush bearing etc.
2. Never try to throttle the suction side of the pump to control discharge as it would seriously affect the performance of the pump.

PROCEDURE

1. Open the delivery valve fully and start the motor.
2. Note the pressure gauge and vacuum gauge readings.
3. Adjust the delivery valve tobring the pressure at 5 kg/cm^2. Note the pressure gauge and vacuum gauge readings.
4. Note the time't' for 5 cm rise of oil in the measuring tank using a stop watch.
5. Also note the time 'T' for 10 impulses of the energy meter to calculate the input power.
6. The above procedure is repeated by gradually decreasing the pressure. (say 4,3,2,1 kg/cm^2)
7. Switch off the power supply.

OBSERVATIONS

Sl No	Time for 5 cm rise of oil (t)	Time for 10 Impulse (T)	Delivery Pressure, h_d		Suction Pressure, h_s		Total Head, H	Actual Discharge Q	Output Power, OP	Input Power, IP	Efficiency
	Sec	Sec	Kgf/cm^2	m	mm of Hg	m	m	m^3/s	kW	kW	%
1											
2											
3											
4											
5											
6											

SAMPLE CALCULATION (Set No :……)

Input power, I.P $= \frac{n\ 3600}{T\ E_m} \eta_m$ kW

Number of imp of energy meter ,n = 10 Imp

Time for 'n' rev of energy meter, T = ………… sec

Energy meter constant, E_m = ……… Imp/kWh

Input power, IP = ………. kW

Efficiency of motor, η_M = 80 %

Output power, OP $= \frac{\rho\ gQH}{1000}$ kW

Acceleration due to gravity, g $= 9.81 \frac{m}{Sec^2}$

Total Head, H = Suction Pressure + Delivery Pressure + Datum Head in m of water

Delivery Pressure head $= \frac{\text{.... } kg/cm2}{Density\ of\ Oil} = \frac{\text{......}}{0.8}$ m

Suction Pressure head (say 20 mm 0f Hg) $= \frac{\text{.......}mm\ of\ Hg}{Density\ of\ Oil} = \frac{\text{......}}{1000\ X\ 0.8}$ 13.6 m of water

Datum Head = 300mm = 0.3 m of water

Total Head, H = ……………………… m of water

Discharge, Q $= \frac{AR}{t} \frac{m^3}{sec}$

Area of collecting tank, A $= 0.25 \times 0.25\ m^2$

Height of water collected, R = 0.05 m

Time for collection, t = …………… sec

Discharge Q $= \frac{AR}{t} \frac{m^3}{Sec} = \frac{0.25\ X\ 0.25\ X\ 0.05}{t} =$ ………… $\frac{m^3}{s}$

Output power OP = ……………. kW

Overall Efficiency $= \frac{Output}{Input} \times 100\ \%$

SAMPLE CALCULATION (Set No :   )

Discharge $\quad Q \quad = \frac{AR}{t}\frac{m^3}{sec}$

Area of collecting tank, A $\quad$ =0.25 x 0.25 = $\quad m^2$

Rise of oil collected, R $\quad$ = 0.05 m

Time for collection, t $\quad$ = sec

Discharge, Q $\quad = \frac{AR}{t} = \quad = \frac{m^3}{sec}$

Output Power, OP, kW $\quad = \frac{\rho_o g Q H}{1000}$

Density of oil, ρ_0 $\quad$ = 800 kg/m^3

Acceleration due to gravity, g $\quad = 9.81 \frac{m}{sec^2}$

Total head, $\mathbf{H = H_d + H_s + l_d}$

H_d= Delivery Pressure head in meters of oil

H_s = Suction Pressure head in meters of oil

l_d=Level difference between delivery and suction pressure headsin meters of oil

h_d - Delivery pressure head, (Kgf/cm^2)

h_s - Suction pressure head, (mm of Hg)

ρ_o- Density of oil, 800 kg/m^3

ρ_{Hg}- Density of Hg, 13600 kg/m^3

Delivery Pressure head $\quad H_d = \frac{h_d}{\rho_o \times 10^{-4}} \quad = \quad =$

Suction Pressure head $\quad H_s = \frac{h_s \times 10^{-3} \times \rho_{Hg}}{\rho_o} = \quad =$

l_d $\quad$ = 300mm $\quad$ = 0.3 m

Total Head, H in meters $\quad H = H_d + H_s + l_d = \quad =$

Output Power, OP, kW $\quad = \frac{\rho_o g Q H}{1000} = \quad =$

Input Power, IP, kW $= \frac{n\ 3600\eta_m}{TE_m}$ =

Number of imp of energy meter, n = 10 Imp
Time for 'n' imp of energy meter, T = sec
Energy meter constant, E_m = Imp/kW h
Efficiency of motor, η_M = 80 %

Input Power, IP,kW $= \frac{n\ 3600\eta_m}{TE_m} =$ =

Overall Efficiency of Gear pump, % $= \frac{OP}{IP} x\ 100 =$

GRAPHS

1. Head v/s Discharge
2. Input Power v/s Discharge
3. Output Power v/s Discharge
4. Efficiency v/s Discharge

RESULT

1. The maximum overall efficiency of the gear pump is %.
2. Required graphs are plotted.

INFERENCE